

Noken genetic algorithms

Gary Nan Tie, Jan 27, 2026

Abstract

Beyond statistical correlation, for the purposes of next-token and sequence-to-sequence token learning, networks of Noken implicit deep learning models with low Renyi α -entropy are sought, in order to sharpen semantic alignment between queries and keys. Genetic algorithms, guided by evolutionary natural selection, are an efficient and systematic way to generate such Noken networks.

Introduction

Through ad hoc trial and error, we could endeavor to configure Noken networks via composition, feedback loops and Rademacher descent, so as to lower resulting Renyi α -entropy, but this is neither systematic nor efficient. Nature however is ruthlessly efficient (and uncaring), a relentless competition of adaptation. Genetic algorithms mimic survival of the fittest to solve optimization problems. We introduce genetic operators specific to Noken implicit models, so as to utilize evolutionary natural selection in designing Noken networks.

References (most recent first)

[1] 'Noken Rademacher Descent'

Gary Nan Tie, Jan 20, 2026

DOI: 10.13140/RG.2.2.20368.47367

Semantic refinement of Noken attention, via query-key similarity alignment, can be achieved by Rademacher descent of implicit model hyperparameters.

[2] 'Semantic Transfer Learning'

Gary Nan Tie, Jan 7, 2026

DOI: 10.13140/RG.2.2.36220.40324

Suppose we have learned the semantics of two different subject domains and want to explore interdisciplinary inquiries. Without starting over from scratch and training on the all combined data (which may in fact muddle matters) we introduce an algorithm to leverage and transfer the individual semantics to make sense of the interdisciplinary corpus.

[3] 'Semantic Attention via Noken Nen'

Gary Nan Tie, Dec 23, 2025

DOI: 10.13140/RG.2.2.10508.99202

Given a target input token from a context string, we can take a draw from a distribution over the string centered at the target, to provide auxiliary contextual information. A Noken, a latent token representing the target, can be potentially refined in a cascade, or iterated in a feedback loop, to utilize the auxiliary information of the drawn token. Convolution like network configurations are explored to reduce average Renyi α -entropy, for the purposes of zero-shot next-token and sequence-to-sequence learning.

[4] 'Noken squared:
Implicit implicit contextual token learning'

Gary Nan Tie, Dec 10, 2025

DOI: 10.13140/RG.2.2.20497.49768/1

'Noken thinking machines'

DOI: 10.13140/RG.2.2.10400.85765/1

Composing implicit token models is a way to update queries in a query-key-value attention mechanism. Coherent refinement is achieved when Renyi α -entropy is reduced. This approach jointly learns query, key, value embeddings and query updates, coordinated by concatenated implicit models.

[5] 'Noken: implicit contextual token learning'

Gary Nan Tie, Dec 4, 2025

DOI: 10.13140/RG.2.2.17440.24329/1

'Noken variations'

DOI: 10.13140/RG.2.2.33204.92805/1

Noken, implicit deep learning models, can be composed into cascades, as well as connected into feedback loops, like Zen Nen. Beyond statistical correlation, Noken networks introduce a novel attention mechanism based on semantic architecture.

[6] 'Implicit Deep Learning'

Laurent El Ghaoui, Fangda Gu, Bertrand Travacca,
Armin Askari, Alicia Y. Tsai

arXiv:1908.06315v4 [cs.LG] 6 Aug 2020

[7] 'Robust Implicit Networks via Non-Euclidean Contractions'

Saber Jafarpour, Alexander Davydov,
Anton V. Proskurnikov and Francesco Bullo

arXiv:2106.03194v6 [cs.LG] 26 Jan 2022

[8] 'Theory of genetic algorithms'

Lothar M. Schmitt

Theoretical Computer Science 259 (2001) 1–61

Token implicit deep learning

For token string $t_1, \dots, t_m \in \mathbb{R}^r$ learn matrices

V, W, Q, K and fixed point $s_i \in \mathbb{R}^n$, $n > r$,

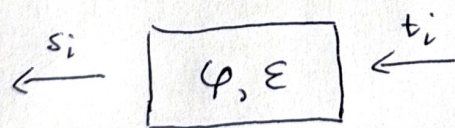
so that for $i \in [m]$:

$$s_i = \varphi(Vs_i + Qt_i)$$

$$\varepsilon = \lambda s_i + Kt_i$$

where V is well-posed for activation function φ

and $\varepsilon \in \mathbb{R}^n$ is a constant vector.



Let $\alpha_{ij} = \text{softmax score}(q_i, x_j)$ where score is a similarity function and $q_i = Q t_i$, $x_j = X s_j$.

Let contextual representation of t_i be:

$$h_i = \sum_{j=1}^m \alpha_{ij} v_j, \text{ where } v_j = V s_j$$

and $\beta_{ij} = \text{softmax score}(h_i, k_j)$, $k_j = K t_j$

Let average Renyi α -entropy of $\{t_i\}$ be:

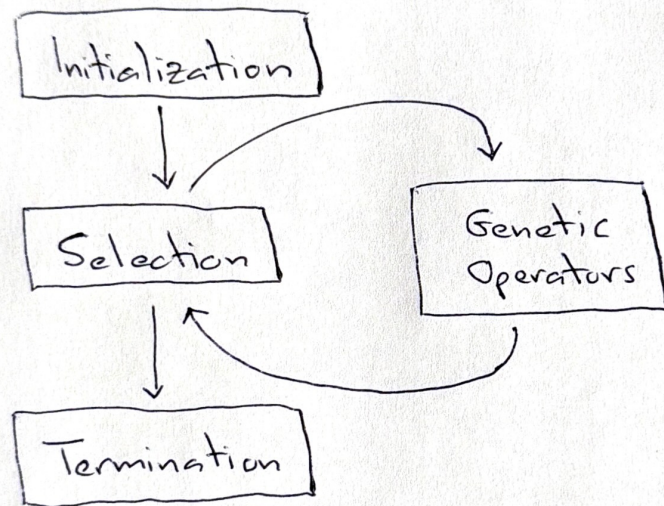
$$E_{\{t_i\}} = E(\{\beta_{ij}\}) = \frac{1}{m} \sum_{i=1}^m H_{\alpha}^i \text{ where}$$

$H_{\alpha}^i = H_{\alpha}(\{\beta_{ij}\}_{j=1}^m)$ and H_{α} is Renyi α -entropy.

We propose using evolutionary algorithms to generate Noken networks for next-token and seq-to-seq learning.

The following generic framework has many variants.

Genetic Algorithm



Step 1: Randomly generate the initial population of individuals.

Step 2: Evaluate the fitness of each individual.

Step 3: Repeat the following regeneration loop until termination:

- i) Select the most fit individuals for reproduction. (Parents)
- ii) Breed children through genetic operators.
- iii) Evaluate the childrens' fitness
- iv) Replace the least fit population with children.

Let an individual be an implicit model $\xleftarrow{s_i} \boxed{\varphi, \varepsilon} \xleftarrow{t_i}$

$$s_i = \varphi(Vs_i + Qt_i)$$

$$\varepsilon = \lambda s_i + Kt_i$$

Draw $\varepsilon \in \mathbb{R}^n$ and

$\varphi \in \{\text{Sigmoid}, \text{tanh}, \text{Maxout}, \text{ReLU}, \text{LeakReLU},$

Exponential Linear Unit $\} =: \Phi$

to randomly generate the first generation

of Noken agents. Iterative selection will build

Noken networks with low Renyi α -entropy.

Fitness for $\xleftarrow{s_i} \boxed{\varphi, \varepsilon} \xleftarrow{t_i}$ is given by $E_{\{t_i\}}$.

Termination depends on the computational cost

of learning implicit models. [See Jafarpour et al, 2022]

Genetic Operators

Mutation Let $\eta > 0$.

Given parent $\xleftarrow{s_i} \boxed{\varphi, \varepsilon} \xleftarrow{t_i}$

draw $\varepsilon' = (\sigma'_1, \sigma'_2, \dots, \sigma'_n)$ where σ_i are independent

with $\text{Prob}(\sigma'_i = \varepsilon + \eta) = \frac{1}{2} = \text{Prob}(\sigma'_i = \varepsilon - \eta)$

Child $\xleftarrow{s'_i} \boxed{\varphi, \varepsilon'} \xleftarrow{t_i}$ has fitness $E_{\{t_i\}}$

Cascade

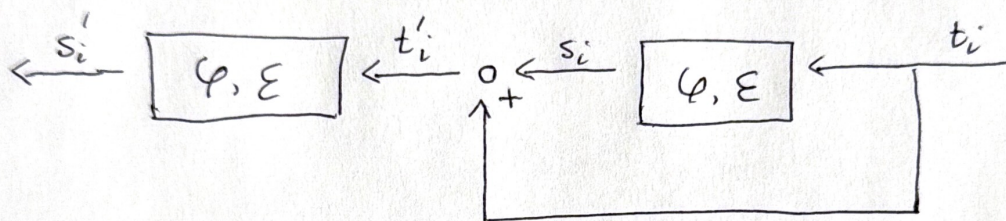
Given parent $\xleftarrow{s_i} \boxed{\varphi, \varepsilon} \xleftarrow{t_i}$ compose as follows:

$\xleftarrow{r_i} \boxed{\varphi, \varepsilon} \xleftarrow{s_i} \boxed{\varphi, \varepsilon} \xleftarrow{t_i}$

Child $\xleftarrow{r_i} \boxed{\varphi, \varepsilon} \xleftarrow{s_i}$ has fitness $E_{\{s_i\}}$

Loop

Given parent $\xleftarrow{s_i} \boxed{\varphi, \varepsilon} \xleftarrow{t_i}$ construct:



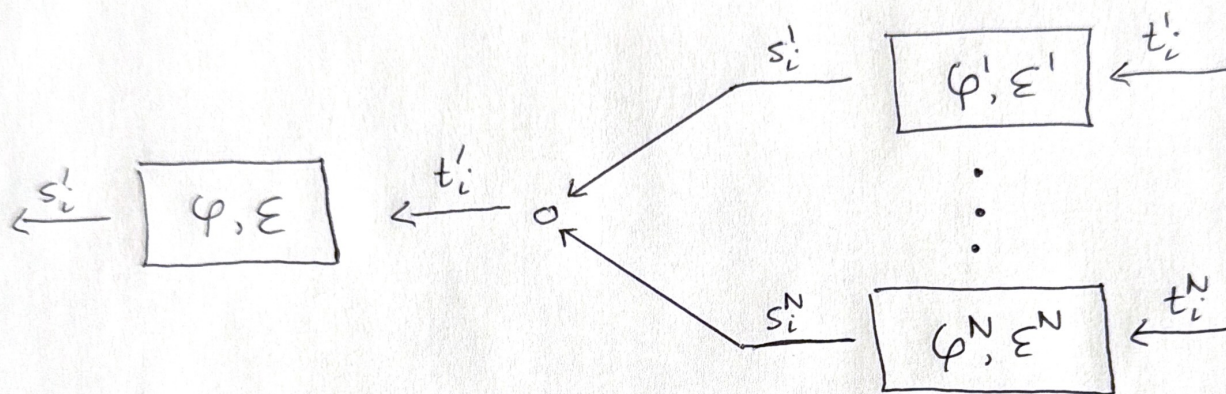
where $t'_i = s_i + t_i$.

Child $\xleftarrow{s'_i} \boxed{\varphi, \varepsilon} \xleftarrow{t'_i}$ has fitness $E_{\{t'_i\}}$

Mix For $k \in [N]$, let* $0 < p_k < 1$ and $\sum_{k=1}^N p_k = 1$.

Choose $\varphi \in \bar{\Phi}$.

Given parents $\xleftarrow{s_i^k} \boxed{\varphi^k, \varepsilon^k} \xleftarrow{t_i^k}$, $k \in [N]$, construct:



where $t'_i = \sum_{k=1}^N p_k s_i^k$ and

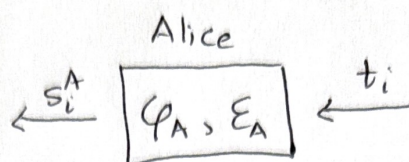
ε is defined by Noken Rademacher descent

starting with $\varepsilon_1 \triangleq \sum_{k=1}^N p_k \varepsilon^k$.

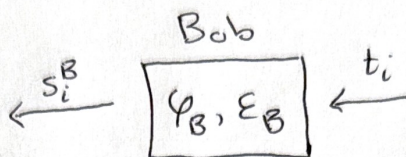
Child $\xleftarrow{s'_i} \boxed{\varphi, \varepsilon} \xleftarrow{t'_i}$ has fitness $E_{\{t'_i\}}$.

* eg $p_k \propto 1/E_{\{t_i^k\}}$

Stack



Parents:



$$t_i \in \{t_1, \dots, t_m\} \subseteq \mathbb{R}^r, \quad \varepsilon_A, \varepsilon_B \in \mathbb{R}^n$$

Alice learns matrices V_A , $\mathcal{H}_A$, Q_A , K_A
 $n \times n$ $n \times n$ $n \times r$ $n \times r$

such that for $i \in [m]$:

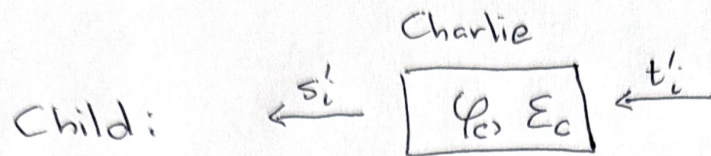
$$s_i^A = \varphi_A (V_A s_i^A + Q_A t_i)$$

$$\varepsilon_A = \mathcal{H}_A s_i^A + K_A t_i$$

where V_A is well-posed for φ_A .

Let E_A denote the average Renyi α -entropy.

Similarly for Bob.



Charlie learns $V, \mathcal{H}, Q, K$ and s'_i so that:

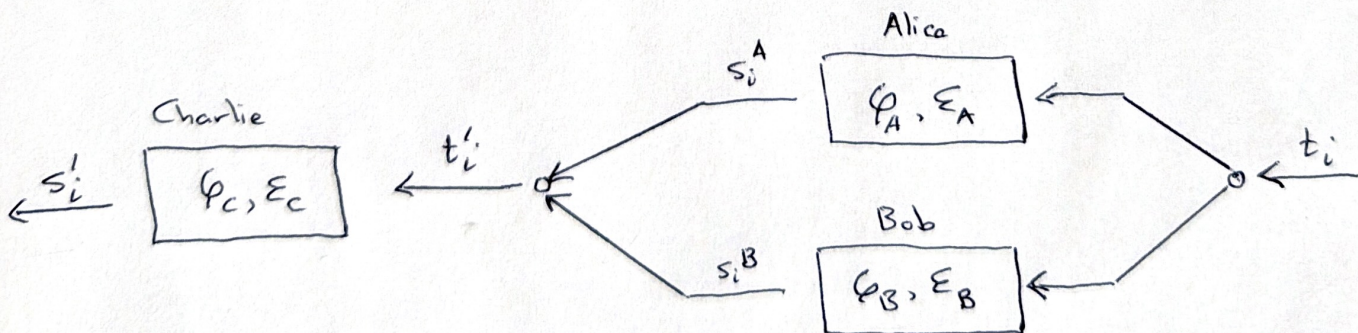
$$s'_i = \varphi_c (V s'_i + Q t'_i)$$

$$\varepsilon_c = \mathcal{H} s'_i + K t'_i$$

where $t'_i = \begin{bmatrix} s_i^A \\ s_i^B \end{bmatrix}$
 2×1

and $\varphi_c, \varepsilon_c = \begin{cases} \varphi_A, \varepsilon_A & \text{if } E_A < E_B \\ \varphi_B, \varepsilon_B & \text{otherwise} \end{cases}$

Charlie has fitness E_c



Noken genetic algorithm

Starting with a randomly generated initial population of Noken implicit models learned on a corpus, we select fit children, bred through the genetic operators mutation, cascade, loop, and mix and stack, to replace the least fit. Here fitness is defined by average Renyi α -entropy of a Noken implicit model. In general, iterating the selection process results in Noken networks with lower entropy, that is more certain semantics in token learning. This Noken genetic algorithm is a form of evolutionary trial and error to build and refine Noken networks.

In Nature there is wisdom, if we look close enough!